Global Perspectives on Water Security:

Water Challenges from The Philippines





Too much

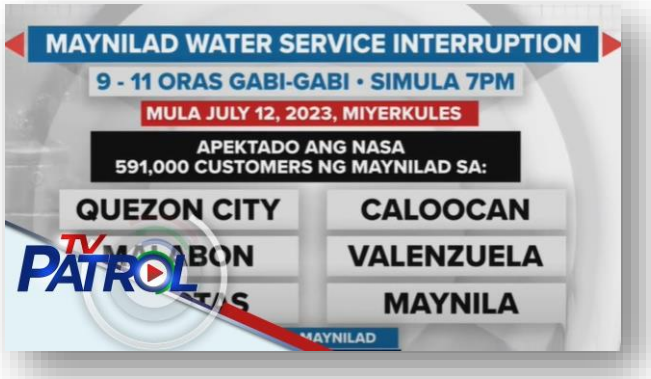



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Too little







Made me wonder and curious about how water system works

Water Security

‘the reliable availability of an acceptable **quantity** and **quality** of water for health, livelihoods, and production coupled with an acceptable level of **water-related risks** to people, environments and econome s ’

- Grey and Sadoff (2007)

“ Security ”

‘ The state of being free from danger or threat ’

But, what is it that makes us secure; and secure from what?

Enough ?

Clean ?

Safe ?

Water Security Dimensions



Rural Household Water Security

– access to safe and inclusive drinking water and sanitation in rural areas



Economic Water Security

– water for agriculture, energy, and industry to support economic productivity



Urban Water Security

– safe, inclusive, and reliable water and sanitation services in urban settings



Environmental Water Security

– protection and restoration of aquatic ecosystems and water-related biodiversity



Water-Related Disaster Security

– capacity to manage and reduce risks from floods, droughts, and storms

Challenge 1: Water Shortage



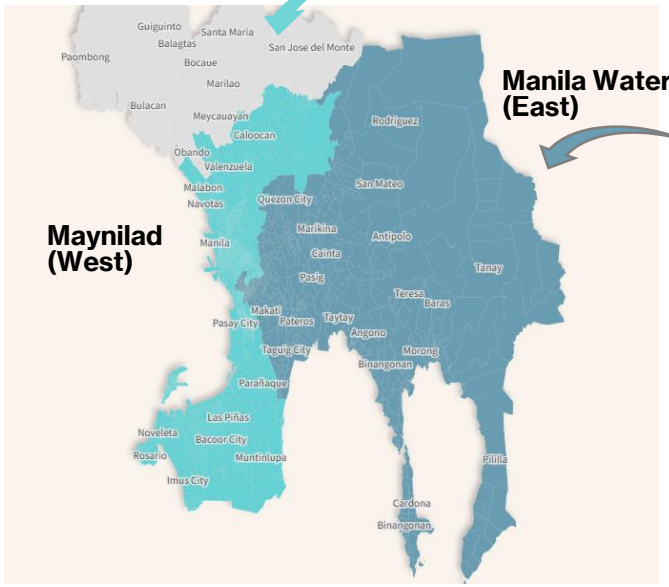


The Philippines will experience a **“high” degree of water shortage by 2040**. It ranked 57th among likely most water stressed country in 2040 out of 167 countries with a 3.01 high water stress level score, meaning, at least 40 percent to 80 percent of the country’s total water supply would have been already withdrawn, or gone, by 2040, with agriculture expected to bear the brunt of the water shortage.

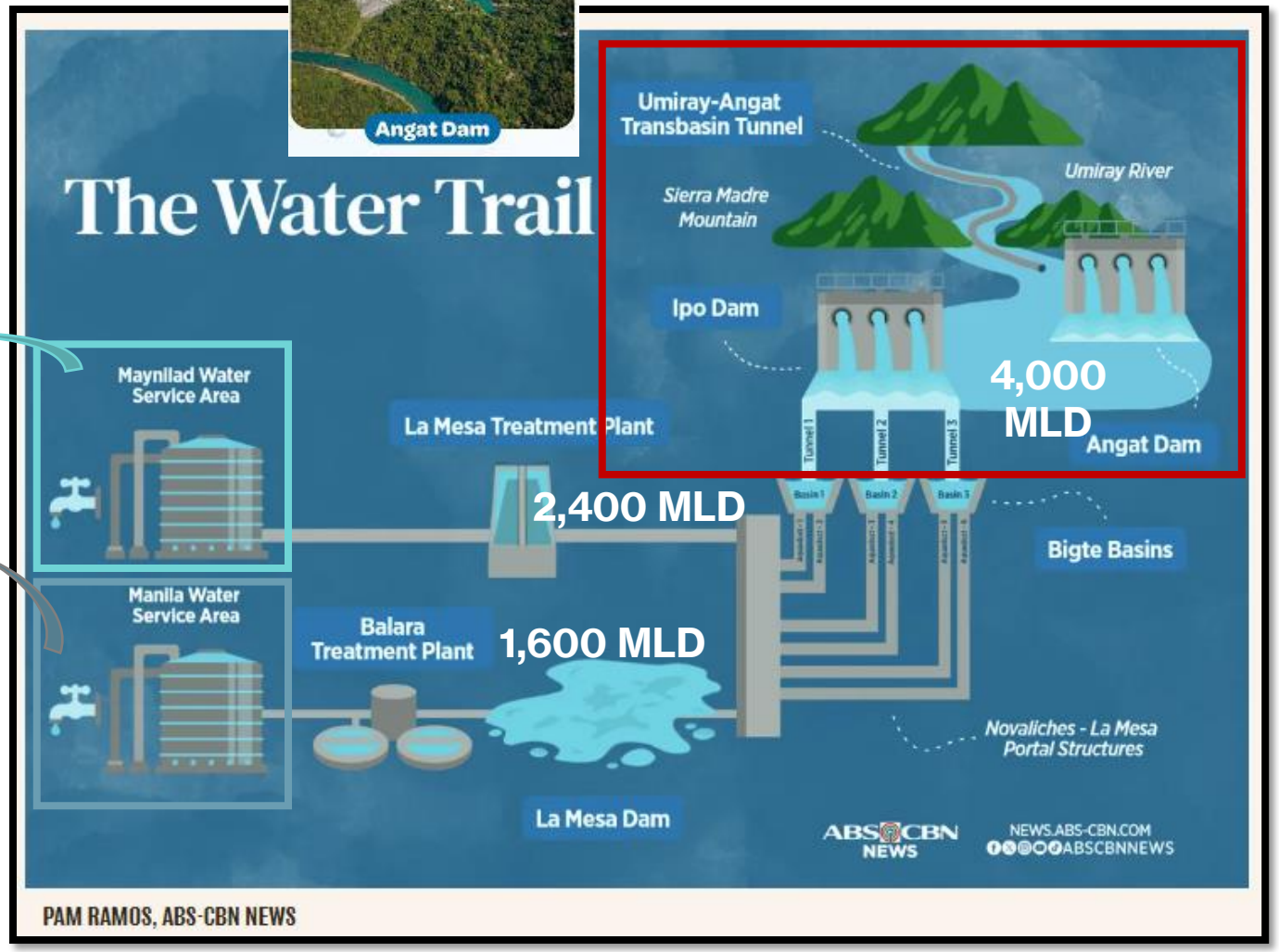


90%

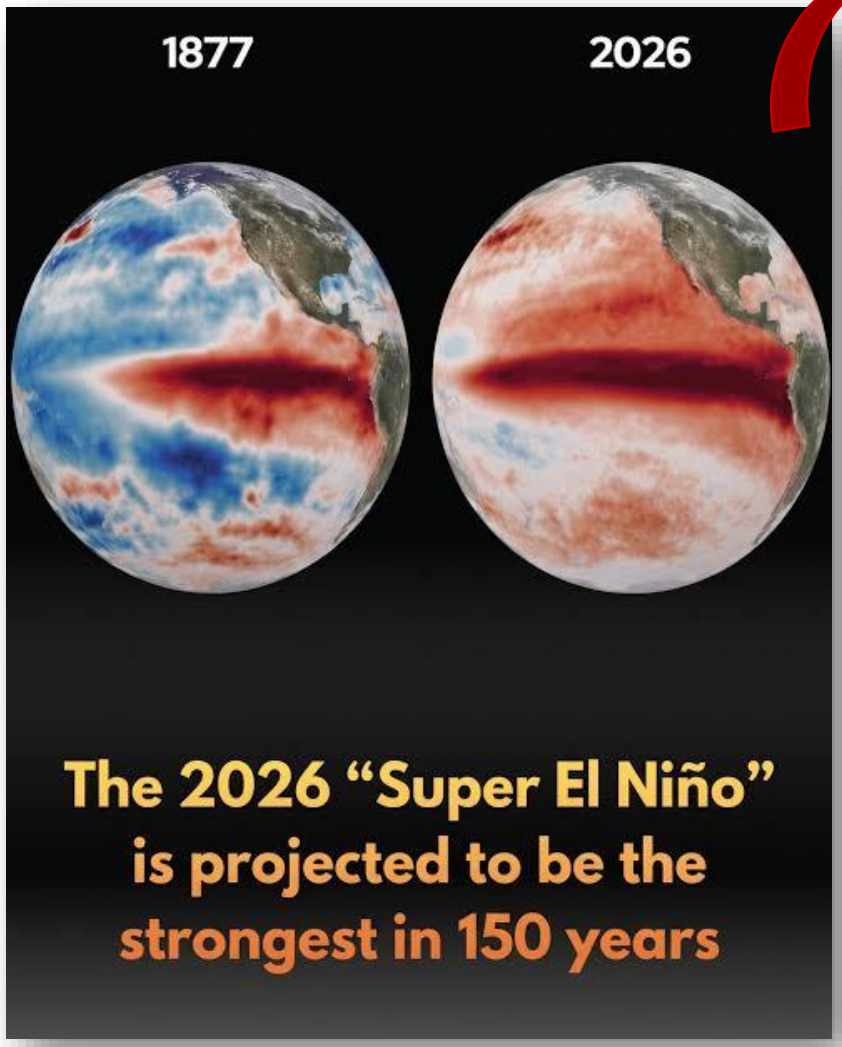
of Manila's water needs come from Angat Dam

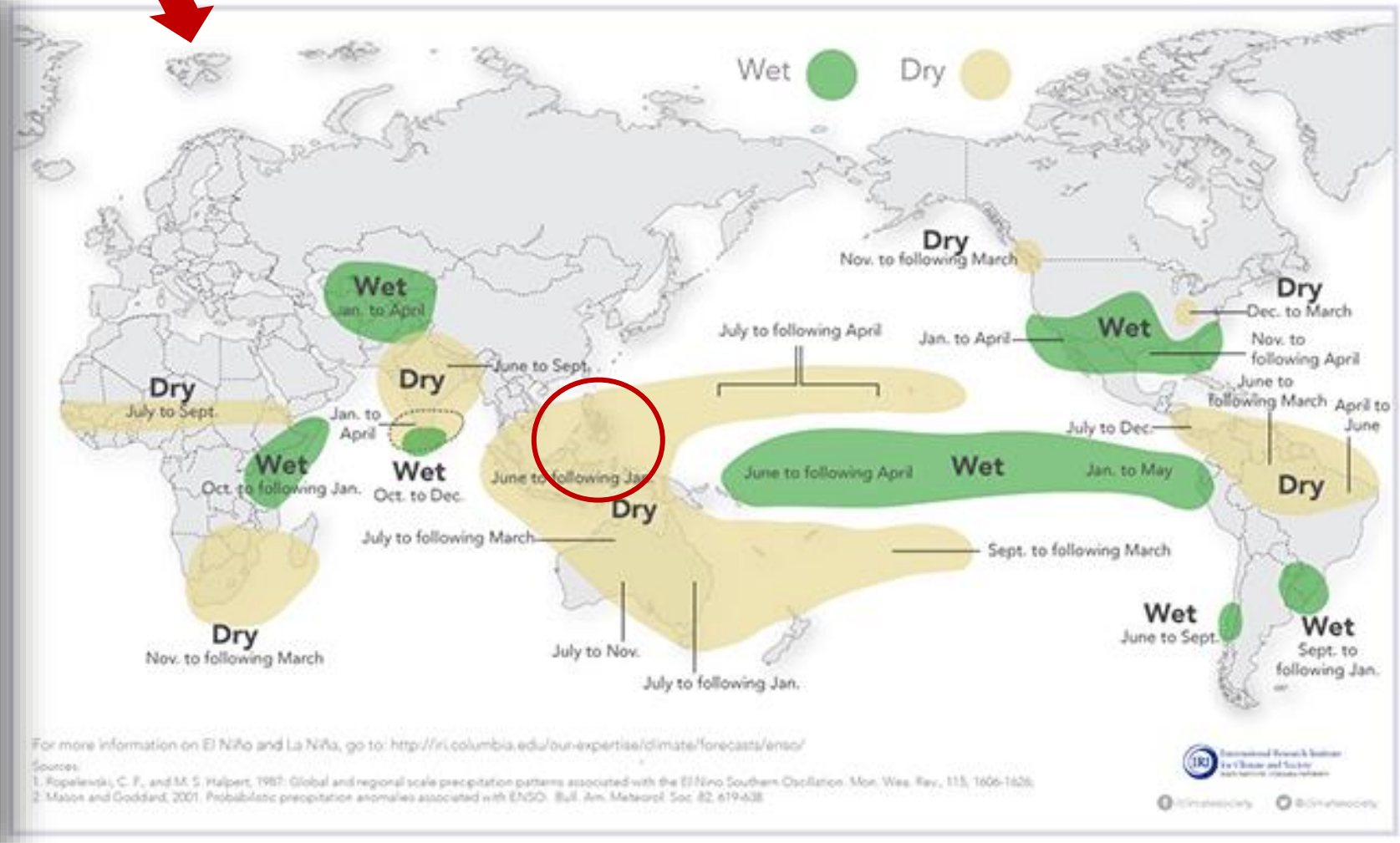


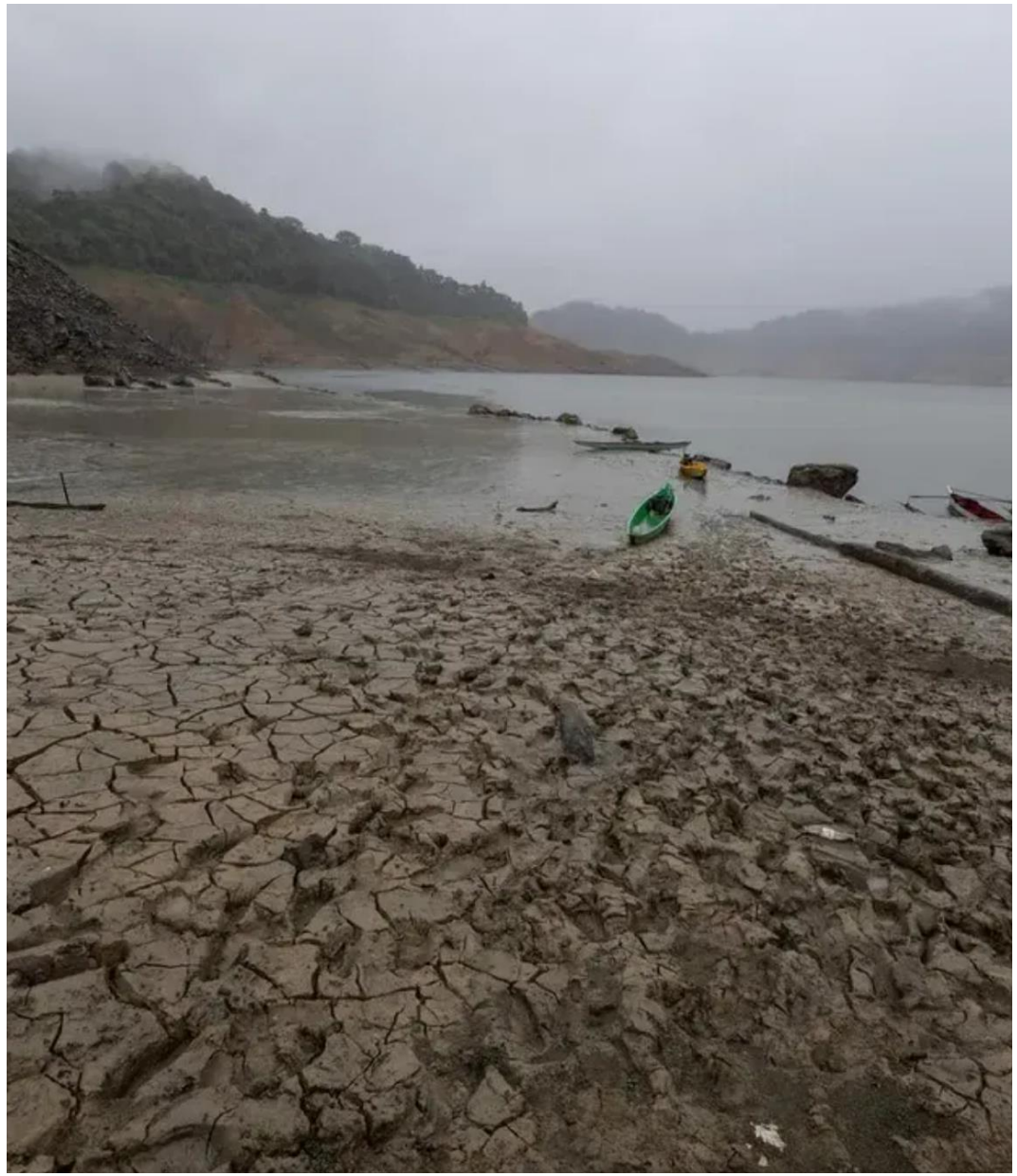




2026 “Super” El Niño







Angat Reservoir During the **June 2026** – Metro Manila, Philippines. Water levels in Angat Dam dipped below the critical mark





In result : **'Budgeting'** Angat Dam supply (Cut water allocation from 53 to 44 m^3/s starting this October 2026)

Top 3 Challenges:

1. Highly dependent on one dominant source
2. Old pipes that lose water
3. Prolonged water infrastructure developments

Challenge 2: Unclean and unsafe







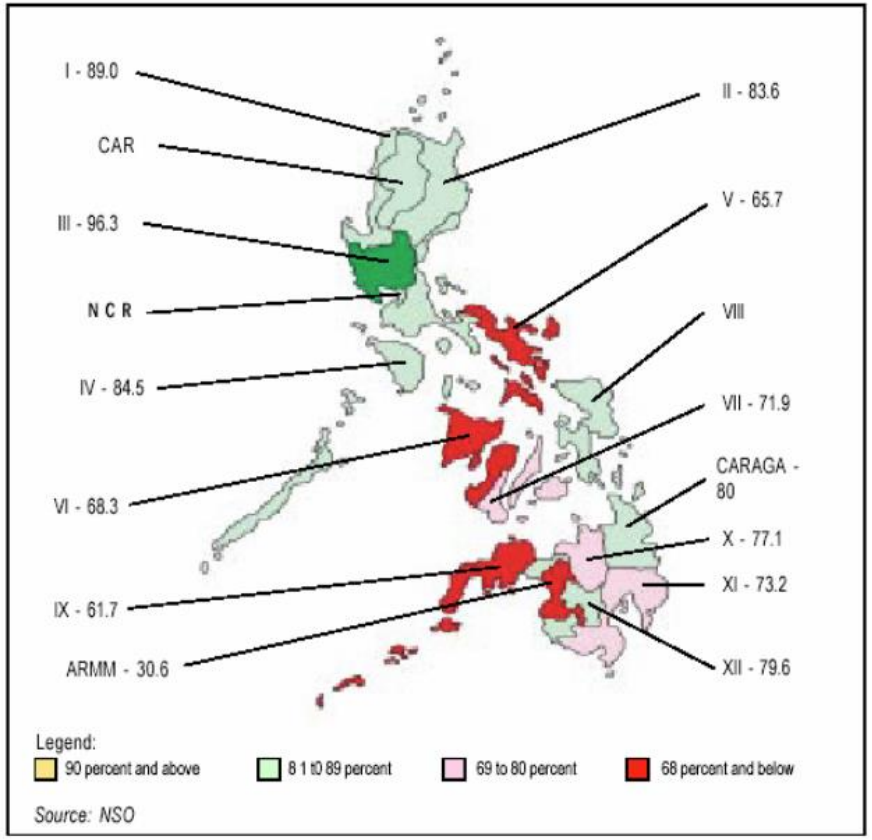
Around **12.4 million Filipinos** still lack access to safe water, and millions more face severe supply shortages and contamination risks.



Quality & Sanitation

51%

of the population lack access to SAFE managed water (59 million people) and 43 million people (37%) lack access to a safe toilet.



Map of families with access to SAFE Water

Top 3 Challenges:

1. In cities, about 62% have safely managed water; in rural areas, only about 35%
2. There are still some untreated sewage that flows into rivers, lakes and groundwater
3. Rivers and lakes are polluted

Environmental



‘Dead River’ **After rehabilitations**







Importance of this:

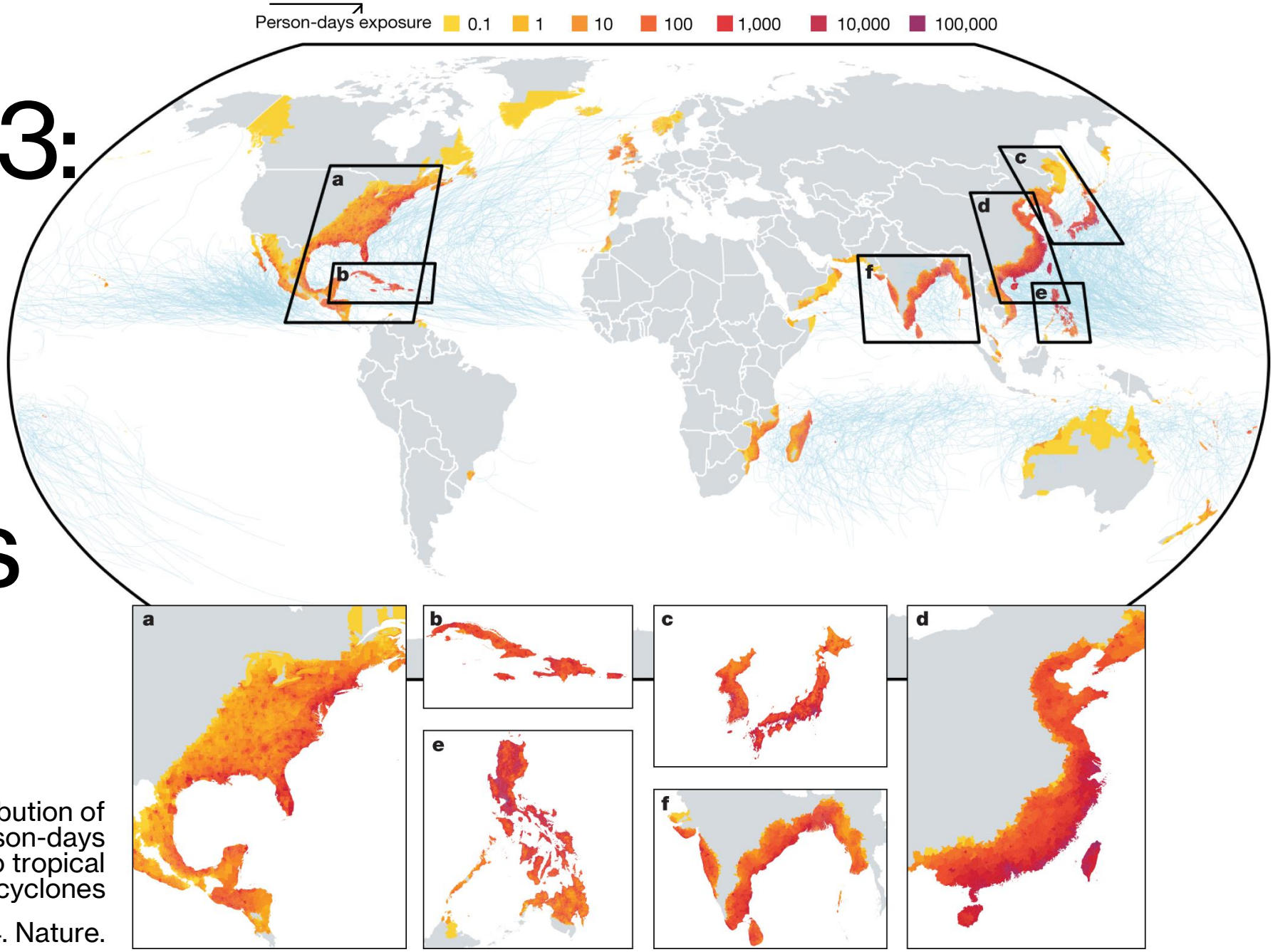
1. Environmental water security is the foundation that supports all other uses of water
2. It acts as climate adaptation – through flood response

Challenge 3: Extreme Weather: Typhoons and Floods



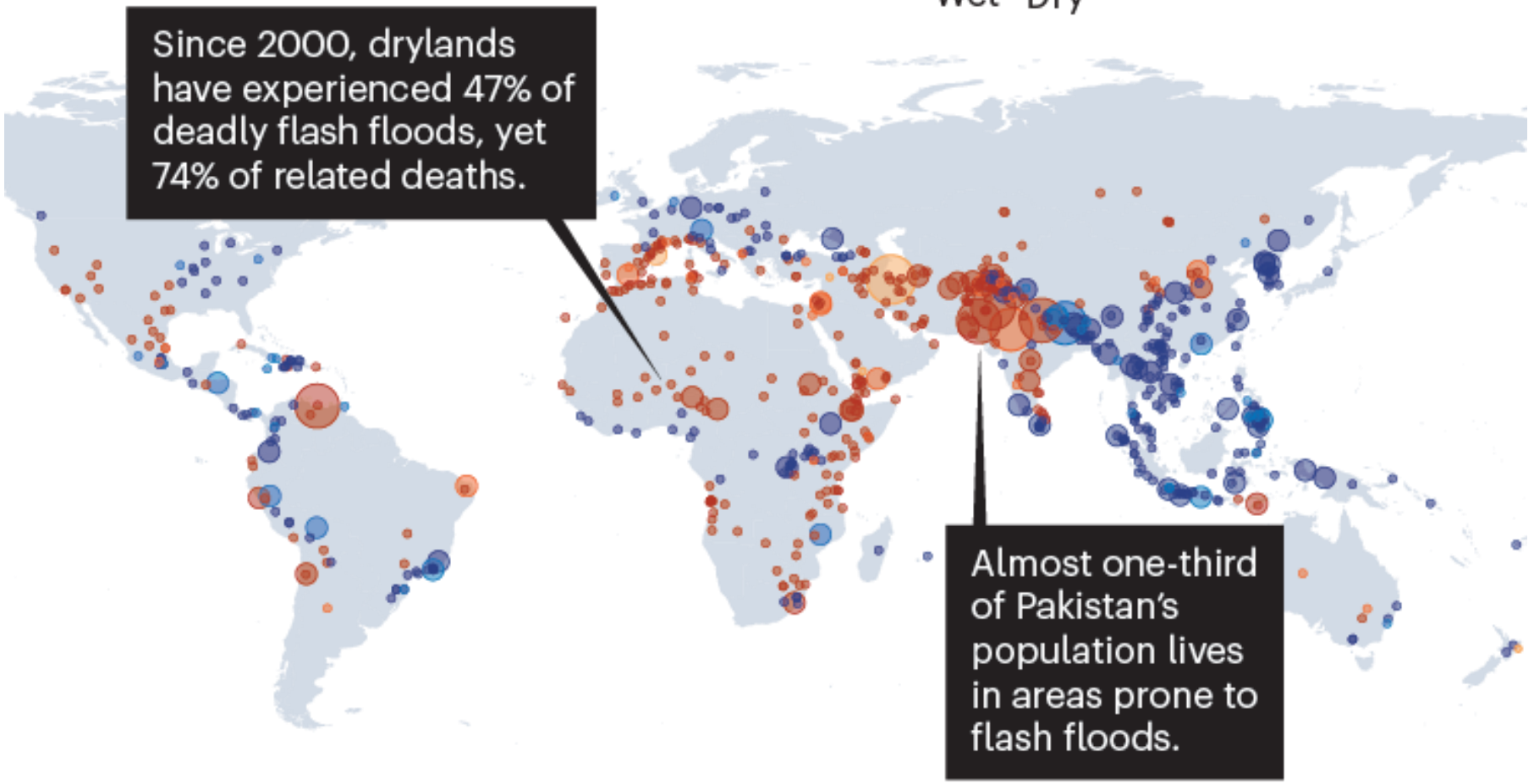
Global distribution of annual person-days exposure to tropical cyclones

Jing et. al. 2024. Nature.



GLOBAL FLASH-FLOOD DISASTERS

Flash floods in drylands kill more people than do those in wetter areas, because dry soils repel water and local populations tend to be unprepared.



*Based on the Köppen climate classification, in which regions other than 'dry' are classed as 'wet'.

©nature

41%

Almost half the world's floods happen in our region

From 2013 to 2023, the Asia Pacific region experienced 244 major floods, 104 droughts, and 101 severe storms.

Climate and water-related disaster risks are rising faster than readiness

Top 3 Challenges:

1. Resilience investments remain fragmented and insufficient
2. Weak institutional capacity and limited integration with local planning and finance hinder effective response
3. Adaptation planning and financing remain slow



Risk to infrastructure destruction

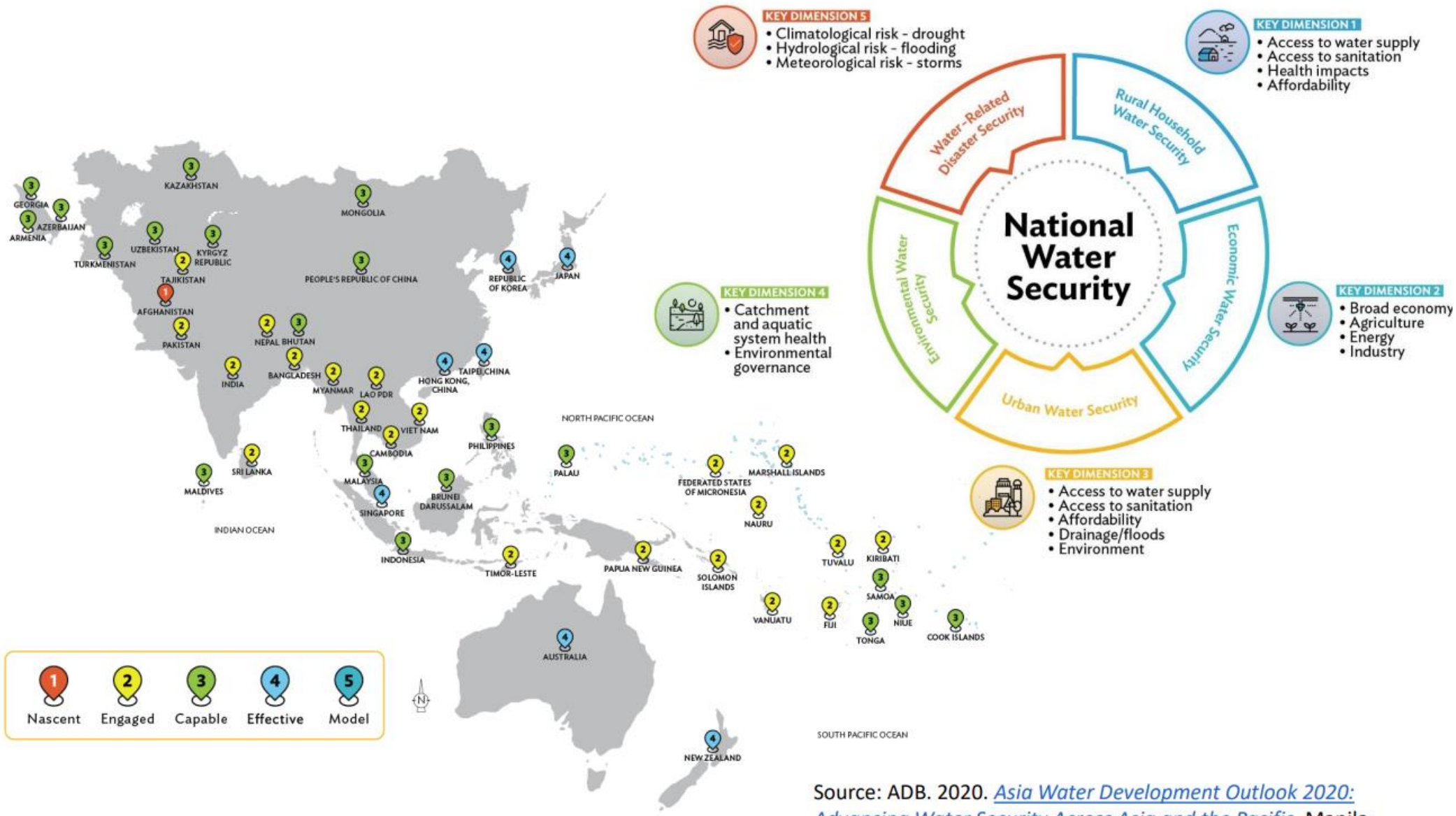


Degrade ecosystems
– reverse development gains



Resource contamination
and Disease outbreaks

State of Water Security in Asia



Source: ADB. 2020. *Asia Water Development Outlook 2020: Advancing Water Security Across Asia and the Pacific*. Manila

Solutions :



Enough Water

- Water conservation
- Water supply assessment
- Storage infrastructures
- Alternative sources
- Improve efficiency and Less waste



Clean Water

- Water treatment infrastructures
- **Water pollution mapping**
- **Ecosystem integrity and biodiversity**
- Enforce water quality regulations
- **Monitoring systems**



Safe Water

- Water sanitation and hygiene
- Water disinfection
- **Early warning systems**
- **Climate-resilient solutions**
- Risk and disaster response
- Climate adaptation preparedness



Salamat.
감사합니다.

6 CLEAN WATER AND SANITATION

13 CLIMATE ACTION